SYLLABUS : MODERN PHYSICS & NUCLEAR PHYSICS

1. Photo electric effect is the phenomenon in which

(A) Photons come out of a metal when it is hit by a beam of electrons

(B) Photons come out of the nucleus of an atom under the action of an electric field

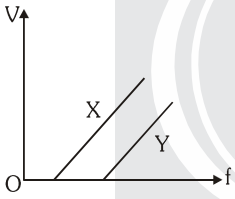
(C) Electrons come out of metal with a constant velocity depending on frequency and intensity of incident light

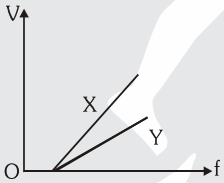
(D) Electrons come out of a metal with different velocity not greater than a certain value which depends only on the frequency of the incident light wave and not on its intensity.

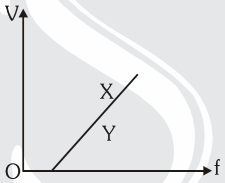
2. Light of two different frequencies whose photons have energies of 1 eV and 2.5 eV are incident one by one on a metal surface of work function 0.5 eV. The ratio of maximum energies of emitted electrons will be

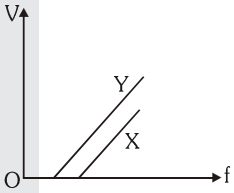
(A) 1 : 4 (B) 4 : 1 (C) 1 : 2 (D) 2 : 1

3. In a photoelectric experiment, electrons are ejected from metals X and Y by light of intensity I and frequency f. The potential difference V required to stop the electrons is measured for various frequencies. If Y has a greater work function than X; which one of the following graphs best illustrates the expected results?

(A) 

(B) 

(C) 

(D) 

4. If the frequency of light in a photoelectric experiment is doubled then maximum kinetic energy of photoelectron

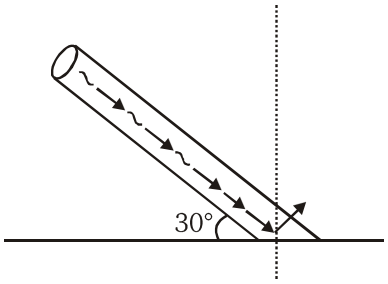
(A) be doubled (B) be halved

(C) become more than double (D) become less than double

5. The maximum kinetic energy of photoelectrons emitted from a surface when photons of energy 6 eV fall on it is 4 eV. The stopping potential is :

(A) 2V (B) 4V (C) 6V (D) 10V

6. The number of photons (λ = 6630Å) that strike per second on a totally reflecting screen (as shown in figure), so that a force of 1N is exerted on the screen, is approximately.



(A) 10^{23} (B) 10^{27} (C) 10^{25} (D) 10^{26}

7. A photon of light enters a block of glass after travelling through vacuum. The energy of the photon on entering the glass block

(A) increases because its associated wavelength decreases

(B) Decreases because the speed of the radiation decreases

(C) Stays the same because the speed of the radiation and the associated wavelength do not change

(D) Stays the same because the frequency of the radiation does not change

8. The de Broglie waves are associated with moving particles. These particle may be–

(A) electrons (B) He^+, Li^{2+} ions (C) Cricket ball (D) All of the above

9. Which of the following statements is wrong ?

(A) De–Broglie waves are probability waves and there is no physical existence of these.

(B) De–Broglie wavelength of a moving particle is inversely proportional to its momentum.

(C) Wave nature is associated with atomic particles only.

(D) In general wave nature of matter is not observed.

10. The wavelength of de–Broglie waves associated with neutrons at room temperature T K is

(A) $\frac{1.82}{T}$Å (B) $\frac{1.82}{\sqrt{T}}$Å (C) $\frac{25.15}{\sqrt{T}}$Å (D) $\frac{30.7}{T}$Å

11. If 13.6 eV energy is required to ionize the hydrogen atom, then the energy required to remove an electron from n = 2 is–

(A) 10.2 eV (B) 0 eV (C) 3.4 eV (D) 6.8 eV

12. In an atom, two electrons move around the nucleus in circular orbits of radii R and 4R. The ratio of the time taken by them to complete one revolution is (neglect electric interaction) :–

(A) 1:4 (B) 4:1 (C) 1:8 (D) 8:1

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13. If a_0 is the Bohr radius, the radius of the n = 2 electronic orbit in triply ionized beryllium is -

(A) $4a_0$ (B) a_0 (C) $a_0/4$ (D) $a_0/16$

14. Which energy state of doubly ionized lithium (Li^{++}) has the same energy as that of the ground state of hydrogen ? Given *Z* for lithium = 3 :

(A) n = 1 (B) n = 2 (C) n = 3 (D) n = 4

15. The wave number of the series limit of Lyman series is–

(A) $1.097 \times 10^7\ m^{-1}$ (B) $2.74 \times 10^6\ m^{-1}$ (C) $1.22 \times 10^6\ m^{-1}$ (D) $6.86 \times 10^5\ m^{-1}$

16. The electron in a hydrogen atom makes transition from M shell to L. The ratio of magnitudes of initial to final centripetal acceleration of the electron is :–

(A) 9:4 (B) 81:16 (C) 4:9 (D) 16:81

17. The ionization energy of hydrogen atom is 13.6 eV. Hydrogen atoms in the ground state are excited by electromagnetic radiation of energy 12.1 eV. How many spectral lines will be emitted by the hydrogen atoms?

(A) one (B) two (C) three (D) four

18. A neutron collides head–on with a stationary hydrogen atom in ground state. Which of the following statements are correct (Assume that the hydrogen atom and neutron has same mass)

(A) If kinetic energy of the neutron is less than 20.4 eV collision must be elastic

(B) If kinetic energy of the neutron is less than 20.4 eV collision may be inelastic

(C) Inelastic collision may be take place only when initial kinetic energy of neutron is greater than 20.4 eV

(D) Perfectly inelastic collision can not take place

19. An electron with kinetic energy 10 eV is incident on a hydrogen atom in its ground state. The collision

(A) must be elastic (B) may be partially elastic

(C) must be completely inelastic (D) may be completely inelastic

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20. The production of characteristic X–rays is due to–

(A) transfer of momentum in collision of electrons with the target atom

(B) transfer of energy in collision of electrons with the target atom

(C) the transition of electrons in heavy target atoms from high to low energy level

(D) none of these

21. According to Einstein's photoelectric equation, the plot of the kinetic energy of the emitted photoelectrons from a metal Vs then frequency, of the incident radiation gives a straight line whose slope :

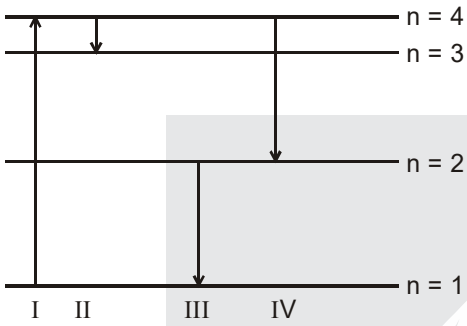
(A) depends on the nature of the metal used

(B) depends on the intensity of the radiation

(C) depends both on the intensity of the radiation and the metal used

(D) is the same for all metals and independent of the intensity of the radiation

22. The diagram shows the energy levels for an electron in a certain atom. Which transition shown represents the emission of a photon with the most energy?



(A) III (B) IV (C) I (D) II

23. If the kinetic energy of a free electron doubles, its de-Broglie wavelength changes by the factor :

(A) $\frac{1}{2}$ (B) 2 (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2}$

24. When ultraviolet light is incident on a photocell, its stopping potential is V_0 and the maximum kinetic energy of the photoelectrons is K_{max}. When X-rays are incident on the same cell, then :

(A) V_0 and K_{max} both increase (B) V_0 and K_{max} both decrease

(C) V_0 increases but K_{max} remains the same (D) K_{max} increases but V_0 remains the same

25. If radiations of allowed wavelengths from ultraviolet to infrared are passed through hydrogen gas at room temperature, absorption lines will be observed in the

(A) Lyman series (B) Balmer series

(C) Both (A) and (B) (D) Neither (A) nor (B)

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ANSWER KEY

1.	(D)	**2.**	(A)	**3.**	(A)	**4.**	(C)	**5.**	(B)
6.	(B)	**7.**	(D)	**8.**	(D)	**9.**	(C)	**10.**	(C)
11.	(C)	**12.**	(C)	**13.**	(B)	**14.**	(C)	**15.**	(A)
16.	(D)	**17.**	(C)	**18.**	(AC)	**19.**	(A)	**20.**	(C)
21.	(D)	**22.**	(A)	**23.**	(C)	**24.**	(A)	**25.**	(A)